Safe Reactive Navigation and Exploration for Planetary Exploration

Matthew Jiang*, Feifei Qian*, Shipeng Liu*†
*University of Southern California, Los Angeles, California 90089, USA
†Corresponding Author
Email: {jiangmy, feifeiqi, shipengl}@usc.edu

***Abstract*—This thesis presents Safe Reactive Planning for Granular Terrain (SRPGT), a navigation framework for autonomous exploration in granular environments where proprioceptive sensing is the main mode of terrain mapping, particularly in scenarios where deformable terrain properties cannot be reliably inferred from visual input. SRPGT enables a legged robot to traverse unfamiliar regolith terrain using only proprioceptive feedback from its limbs.**

The core contribution is a dual-layer architecture combining global terrain expansion with real-time reactive control. The global layer models terrain risk using a Gaussian Process updated through in-place mechanical tests at each step. It incrementally expands a certified safe set of traversable regions using a confidence-aware exploration strategy inspired by Safe Bayesian Optimization. The local control layer uses a diffeomorphism-based reactive controller adapted from power diagram methods, allowing the robot to navigate through the self-generated concave environments without requiring global replanning.

SRPGT is shown to successfully balance exploration and exploitation, avoid unsafe regions, and can generalize across navigation and pure exploration tasks. The algorithm achieves real-time performance and robustness without vision.

I. Introduction

Planetary exploration increasingly demands autonomous robotic systems capable of navigating hazardous terrain without direct human supervision. Traditional wheeled rovers rely heavily on vision-based sensing to interpret their environment. The Mars 2020 Perseverance rover uses Hazard Avoidance Cameras (HazCams) and Navigation Cameras (NavCams) to map terrain hazards [10], but this approach presents significant limitations when operating in granular, deformable terrains common on planetary surfaces. Materials such as regolith, volcanic ash, and fine particulate dust can be deceptive: two visually identical patches may respond differently under load, with one providing stable support while the other collapsing under minimal pressure.

The consequences of these limitations are well documented. In 2009, the Spirit rover became irretrievably stuck in Martian sand despite careful visual navigation, and a tragic comic was published [12], highlighting the fundamental inadequacy of existing methods, such as vision, for assessing terrain mechanical properties. Cameras, regardless of sophistication, cannot directly measure critical characteristics such as cohesion, bearing strength, or subsurface instability that determine traversability in granular media.

These challenges motivate a shift toward active physical interrogation of terrain. Legged robots offer a unique advantage for such interactions, as each step becomes an opportunity to measure mechanical response through proprioceptive sensing. By applying controlled forces and recording resulting displacements, legged platforms can directly infer terrain properties critical for safe navigation, unavailable through vision alone.

This work presents a method that combines safe Bayesian global planning with geometry-based local control for granular terrain exploration. The system enables a legged robot to actively map terrain mechanical properties through proprioceptive interaction, update risk estimates in real time, and select both destinations and safe paths while maintaining operational safety and supporting scientific objectives.

II. Related Work

Autonomous navigation in unstructured environments has been addressed through two primary approaches: probabilistic terrain modeling for safe exploration and reactive control for real-time obstacle avoidance. However, existing methods predominantly rely on exteroceptive sensing, limiting their applicability in visually degraded environments common in planetary exploration.

A. Gaussian Process-Based Terrain Navigation

Bayesian optimization frameworks using Gaussian Processes (GPs) have emerged as powerful tools for terrain-aware navigation under uncertainty. Muenprasitivej et al. [11] employ GPs to model terrain elevation and uncertainty for bipedal robots, integrating footstep planning with information gain objectives. Uttsha et al. [18] extend this by creating GP-based distance fields from 3D point clouds, enabling robust trajectory optimization for both legged and wheeled platforms. Leininger et al. [8] combine Sparse GPs with RRT* planning, selecting subgoals to minimize exposure to uncertain terrain regions. Stephens et al. [16] further apply GPs to encode safety-relevant terrain features within a Markov Decision Process framework, enabling safe exploration in sparse and hazardous environments.

While these approaches demonstrate the effectiveness of probabilistic terrain modeling, they face significant limitations for real-time operation: they require exteroceptive sensors (cameras, LiDAR), rely on global map construction whose computational complexity scales with environment size, and depend on batch planning methods like RRT* that become prohibitively expensive as the explored area grows.

B. Classical Terrain-Aware Planning Methods

Classical path planners remain essential for terrain navigation due to their theoretical guarantees and algorithmic flexibility. A* and its dynamic variants (e.g., D*, D* Lite) support efficient replanning in partially-known environments, using grid-based terrain cost maps that reflect slope, roughness, or stability. For example, Stentz [15] introduce D*, enabling robots to incrementally update their shortest path as new terrain information becomes available.

Sampling-based methods such as RRT and its optimal variant RRT* allow navigation in high-dimensional or continuous spaces. Karaman and Frazzoli [7] demonstrate that RRT* can asymptotically converge to minimum-cost paths, a useful property when terrain traversability costs are integrated. Jaillet et al. [6] extend this with T-RRT, which guides tree growth through low-cost, more traversable regions using a transition-based acceptance criterion.

RRT-X, introduced by Otte and Frazzoli [13], is an asymptotically optimal, single-query sampling-based motion planning algorithm designed for real-time navigation in dynamic environments. It maintains and incrementally repairs a search graph as the robot moves and as obstacles appear or disappear. While RRT-X offers rapid replanning capabilities, it still requires an initial global plan and its computational complexity scales with the size of the environment, which can be limiting in large or highly dynamic terrains.

These planners offer strong global coverage and robustness, but like GP-based approaches, they often require terrain cost maps derived from exteroceptive data and incur high computational cost when the environment size grows or when terrain varies at fine spatial resolution.

C. Reactive Navigation and Voronoi Methods

Complementing probabilistic approaches, reactive navigation methods enable real-time collision avoidance as new obstacles are discovered. Arslan and Koditschek [1] propose using power diagrams to define convex obstacle-free regions, with local optimization guiding robots toward goals with convergence guarantees. Arslan and Koditschek [2] extend this to unknown environments using local sensing to infer separating hyperplanes for reactive control fields.

Recent work by Vasilopoulos et al. [19] and Vasilopoulos et al. [20] combines semantic SLAM with reactive control, enabling smooth motion in dynamic environments through object detection and scene understanding. Garrido and Moreno [5] use Voronoi diagrams with Fast Marching Methods for maximal clearance path planning with continuous obstacle updates.

These reactive methods achieve impressive real-time performance with computational complexity independent of map size, but depend on explicit obstacle detection through vision or range sensors, limiting their effectiveness when exteroceptive sensing is degraded or unavailable.

D. Knowledge Gap and Contribution

Existing literature reveals that probabilistic terrain methods excel at uncertainty-aware exploration but suffer from computational scaling issues with map size and require exteroceptive sensing, while reactive methods enable real-time navigation with constant computational complexity but depend on explicit obstacle detection. Classical planning algorithms offer global path optimality or completeness, yet similarly rely on prior cost maps and scale poorly in dense or dynamic terrain.

No prior work addresses real-time navigation in granular, deformable terrain using proprioceptive sensing.

This work bridges this gap by integrating GP-based uncertainty modeling with reactive control in a computationally efficient framework that maintains constant-time navigation performance regardless of explored area size. Unlike previous methods that model terrain elevation or detect geometric obstacles, our approach directly estimates mechanical terrain properties through physical interaction, enabling real-time safe exploration in visually degraded environments.

III. Methods

A. Terrain-Aware Navigation as a Sequential Decision Problem

Legged robots are uniquely suited to granular terrain because their limbs can serve two functions: enabling mobility and acting as embedded sensors. Each step provides an opportunity for measurement and becomes an input to an evolving internal model of terrain risk.

Navigation in this context can be treated as a sequential decision-making process under uncertainty. The robot must choose its next move based on local terrain observations, estimated risks, and proximity to the goal, while ensuring that all movement adheres to strict safety constraints. This challenge is met by combining two core components:

1) A probabilistic planner that incrementally expands a certified safe set of traversable terrain using terrain observations and confidence-based reasoning.
2) A reactive controller based on diffeomorphic transformation that enables real-time, geometry-aware path execution within certified safe regions.

B. Problem Statement

The robot operates within a two-dimensional discretized domain $D \subset \mathbb{R}^2$, where each point has an unknown terrain property, such as shear strength or cohesion, that influences traversability. The objective is to move from an initial location x_0 to a goal location x_g, ensuring that all intermediate positions lie within regions classified as safely traversable.

We define $f(x)$ as a scalar-valued function representing the traversability of terrain at location x, where higher values correspond to more easily traversable regions. This function is treated abstractly and may encode various underlying physical or empirical terrain characteristics depending on the application context. A fixed threshold h is used to determine safety: terrain is considered safe if $f(x) \geq h$, and unsafe if $f(x) < h$.

The values of $f(x)$ and h are not assumed to carry physical units or universal semantics; rather, they serve as internally consistent constructs for distinguishing safe and unsafe terrain within the chosen representation framework.

Key assumptions and constraints are as follows:

- Terrain risk is a continuous but unknown function over space.
- Proprioceptive terrain measurements are available only at the robot's current location.
- Navigation must maintain high-confidence safety guarantees at all times.

C. Confidence-Guided Terrain Expansion

The planner maintains a probabilistic model of terrain risk using a Gaussian Process (GP). At each step, it updates terrain predictions and refines a subset of the environment certified as safe.

This method draws inspiration from prior work on safe exploration algorithms such as SafeOPT and SafeUCB [17, 9], but differs in structure and execution. In particular, the focus of this work is not on maximizing a reward function, but on expanding reachable terrain while ensuring probabilistic safety at every decision point.

1) Gaussian Process Terrain Modeling: The robot uses proprioceptive data to train a Gaussian Process (GP) model over D, producing a mean prediction $\mu_t(x)$ and standard deviation $\sigma_t(x)$ at each location $x \in D$. The GP uses a radial basis function (RBF) kernel:

$$k(x, x') = \sigma^2 \exp\left(-\frac{\|x - x'\|^2}{2\ell^2}\right)$$

where σ^2 is the kernel variance and ℓ is the lengthscale, both selected manually based on the expected variability of the proprioceptive signal. These hyperparameters are fixed at deployment time.

The model is updated recursively using all past measurements for computational efficiency. At each step, the robot collects a new proprioceptive terrain measurement y_t at position x_t, then incorporates this into the GP model to update posterior predictions across the domain.

2) Safe Set Expansion: A fixed exploration parameter β_t is used to control the width of confidence intervals:

$$Q_t(x) = \left[\mu_{t-1}(x) \pm \sqrt{\beta_t}\,\sigma_{t-1}(x)\right] \tag{1}$$

This formulation is adapted from the SafeOPT algorithm introduced by [17], where confidence intervals are used to balance safety with informative sampling. The value of β_t should be chosen via empirical tuning to balance conservative exploration with steady progress.

The resulting confidence bounds are intersected with prior confidence sets to form updated constraints:

$$C_t(x) = C_{t-1}(x) \cap Q_{t-1}(x) \tag{2}$$

From these sets, the lower and upper bounds are defined as:

$$\ell_t(x) = \min C_t(x), \quad u_t(x) = \max C_t(x) \tag{3}$$

and the interval width is computed as:

$$w_t(x) = u_t(x) - \ell_t(x) \tag{4}$$

Algorithm 1 Safe Optimization for 2D Navigation

Require: Sample set D, GP prior (μ_0, k, σ_0), Lipschitz constant L, seed set S_0, safety threshold h, goal location x_g, number of expanders to consider n

$C_0(x) \leftarrow [h, \infty)$, $\forall x \in S_0$
$C_0(x) \leftarrow \mathbb{R}$, $\forall x \in D \setminus S_0$
$Q_0(x) \leftarrow \mathbb{R}$, $\forall x \in D$
for each time step $t = 1, 2, \ldots$ **do**
  $C_t(x) \leftarrow C_{t-1}(x) \cap Q_{t-1}(x)$
  $S_t \leftarrow \bigcup_{x \in S_{t-1}} \{x' \in D \mid \ell_t(x) - L\|x - x'\| \geq h\}$
  $G_t \leftarrow \{x \in S_t \mid g_t(x) > 0\}$
  Sort G_t by $\|x - x_g\|$ in ascending order
  Let $G_t^{(n)} \subseteq G_t$ be the top n nearest expanders
  $x_t \leftarrow \arg\max_{x \in G_t^{(n)}} w_t(x)$
  Observe $y_t = f(x_t) + n_t$
  Compute $Q_t(x)$ for all $x \in S_t$
end for

To identify G_t, the function

$$g_t(x) = |\{x' \in D \setminus S_t \mid u_t(x) - L\|x - x'\| \geq h\}| \tag{5}$$

is defined, evaluating the expansion potential of each point.

This strategy combines uncertainty-driven sampling with spatial prioritization to ensure that each move both refines the terrain model and progresses toward the mission objective.

The robot selects the next point to sample by balancing two objectives:

- Reducing model uncertainty (exploration)
- Making progress toward the mission goal (exploitation)

After constructing the safe set S_t and identifying the candidate expanders $G_t \subseteq S_t$, the robot must select the next point at which to sample terrain. Rather than choosing purely based on uncertainty or purely based on proximity to the goal, SRPGT employs a hybrid strategy to guide exploration in a goal-directed yet uncertainty-aware manner.

3) Next Parameter Selection: The sub-goal selection procedure is as follows:

1) For each point $x \in G_t$, compute its distance to the goal $\|x - x_g\|$.
2) Sort all points in G_t in ascending order of distance to the goal.
3) Consider only the top n nearest expanders, where n is a configurable parameter.
4) Among these n expanders, select the point with the largest confidence interval width $w_t(x)$.

This approach prioritizes expanders that offer informative terrain data, while guiding the robot towards its goal by using a distance heuristic. It mitigates the risk of wasting samples on irrelevant or backward directions and prevents purely greedy selection based on proximity alone, which might lead into regions of high uncertainty without expanding the safe set.

By focusing on a small subset of candidate expanders that are both near the goal and informative, the planner ensures steady, efficient progress toward the goal while incrementally enlarging the verified safe zone.

D. Reactive Voronoi-Based Navigation

While the planner identifies the next point to move toward, executing safe and efficient motion between steps requires a robust local navigation strategy. Classical global planners, such as A* and RRT*, require full replanning after significant environmental updates, making them unsuitable for highly dynamic, partially known terrains.

Instead, a reactive control framework inspired by [20] is implemented.

1) Synthesizing Pseudo-Physical Obstacles: To support reactive navigation, the robot must construct geometric representations of both safe and hazardous terrain regions from its evolving internal risk map. This process transforms pointwise risk predictions into polygonal approximations of safe space and its complement, enabling geometry-aware motion planning.

The environment is discretized into a two-dimensional grid. In simulation, this grid is aligned with the resolution of the ground truth map for consistency. In real-world deployments, however, the resolution becomes a tunable parameter. Higher resolutions enable more detailed boundary tracking but increase computational overhead, while lower resolutions reduce map fidelity in favor of speed and memory efficiency.

The conversion from safe points to obstacle geometry proceeds as follows:

1) **Cluster Formation:** The current safe set S_t is partitioned into disjoint spatial clusters $\{S_{ti}\}$ using a combination of KD-tree nearest-neighbor queries and union-find data structures. Each cluster represents a contiguous region of confidently traversable terrain.
2) **Concave Hull Construction:** For each cluster S_{ti}, a concave hull H_{ti} is generated using the algorithm described by [14] and shown in Figure 1, which takes as parameters a relative concavity level $\mathcal{C}$ and a local threshold L_{th}. These settings determine the hull's shape resolution and computational complexity. In our implementation, we select high concavity values to tightly wrap each cluster. While these parameters influence geometric fidelity, they do not affect overall safety guarantees and may be tuned based on application requirements.
3) **Workspace Definition:** A bounding rectangle $\mathcal{W}_e$ is formed to enclose all H_{ti} regions, as depicted in Figure 2:
$$\mathcal{W}_e := \left\{x \in \mathbb{R}^2 \,\middle|\, \underline{H}_t \leq x \leq \overline{H}_t\right\} \tag{6}$$
where $\underline{H}_t = [\min_i H_{ti}^{(1)}, \min_i H_{ti}^{(2)}]^T$ and $\overline{H}_t = [\max_i H_{ti}^{(1)}, \max_i H_{ti}^{(2)}]^T$.
4) **Obstacle Extraction:** The regions not covered by the union of safe hulls are treated as pseudo-physical obstacles:
$$\mathcal{O}_t := \mathcal{W}_e \setminus \bigcup_i H_{ti} \tag{7}$$
5) **Polygon Simplification:** The obstacle set $\mathcal{O}_t$ is polygonal by construction but may contain excessive vertices. Each obstacle polygon is simplified using the Douglas-Peucker algorithm to reduce computational cost while preserving topology for subsequent triangulation and transformation.

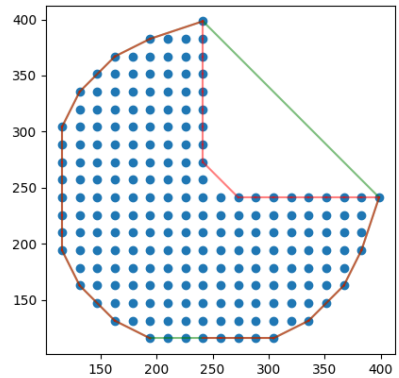

Fig. 1. Visualization of a concave hull generated from point clusters. Image adapted from the `cubao/concave_hull` GitHub repository [4]. Licensed under the MIT License.

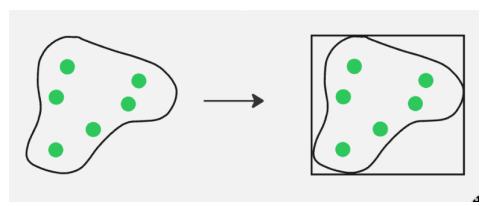

Fig. 2. Visualization of bounding box creation. Green dots are fictional locations of safe samples, inner shape is depiction of safe area.

The resulting polygon set $\mathcal{O}_t$ is used by the reactive controller to generate safe, smooth navigation commands around terrain hazards that have been inferred from contact-based measurements.

2) Diffeomorphic Mapping and Reactive Control: For local motion within the safe set, we adopt the reactive navigation framework proposed by Vasilopoulos et al. [20]. Their method constructs a diffeomorphic transformation between the physical workspace and a geometrically simplified model space, enabling obstacle avoidance and convergence to the goal using a simple potential field in the transformed domain.

The approach guarantees that smooth trajectories in the model space correspond to safe, obstacle-free paths in the physical space, provided the obstacle set is polygonal. This structure allows the robot to reactively generate control inputs in real time without full replanning, even as the set of known obstacles evolves.

We apply this framework directly to the polygonal obstacle

set derived from proprioceptive terrain assessments, treating high-risk or unobserved terrain as non-traversable. The resulting controller ensures that all local motions remain within the verified safe region while making progress toward the current subgoal.

This adaptation preserves the convergence and safety guarantees of the original method while extending its applicability to environments where obstacles are not visually detected but inferred from mechanical interaction.

E. System Integration

The final system integrates global terrain expansion and local reactive control into a unified exploration loop:

- Proprioceptive measurements are gathered during motion.
- The GP-based model of terrain risk is updated after each measurement.
- The planner selects a new target location from safe or candidate expander sets.
- The reactive controller executes real-time movement toward the selected target.

This dual-loop architecture allows the robot to safely explore unknown, deformable environments while efficiently progressing toward its scientific objectives.

IV. Results

This section presents simulation-based validation of SRPGT across terrain scenarios that demonstrate its core capabilities: safe region expansion under uncertainty, navigation around high-risk areas, reuse of learned terrain knowledge, and exploratory mapping without preassigned goals.

A. Simulation Setup

The robot is initialized within a verified safe region with randomly sampled proprioceptive measurements to bootstrap the Gaussian Process model. Test environments are represented as 2D grids where each cell contains terrain risk values, with the traversability function $f(x)$ encoding relative terrain strength and safety threshold h defining risk boundaries (e.g., $h = 1000$ for demonstration).

Visualization Legend: White = safe terrain ($f(x) \geq h$); Gray = unsafe terrain ($f(x) < h$); Black = obstacles ($\mathcal{O}_t$); Green = local freespace; Light Blue = safe set (S_t); Red = expanders (G_t); Pink = intermediate goal; Dark Blue = final goal.

B. Experimental Results

a) General Navigation: When the direct path contains no risk zones, the robot efficiently expands the safe zone and reaches the goal in 54 iterations (Figure 4).

b) Navigation Around Risk Zones: The robot successfully navigates around high-risk regions by iteratively expanding the safe zone until reaching goals beyond obstacles. Two scenarios completed in 210 and 461 iterations respectively (Figure 5, Figure 6). The algorithm initially expands rapidly, then explores cautiously around risk peripheries, and finally accelerates once past dangerous areas.

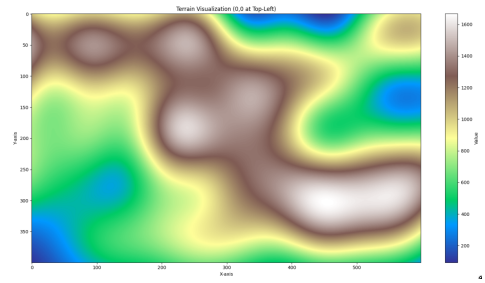

Fig. 3. Ground truth risk values for experiments, starting at $(230, 220)$, goal at $(104, 82)$.

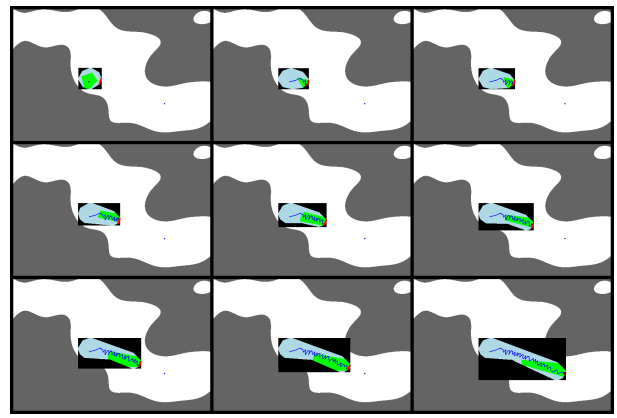

Fig. 4. Simple navigation task progression over 9 evenly spaced intervals.

c) Knowledge Reuse: After establishing a large safe area, the robot efficiently traverses previously learned safe corridors. Navigation across the environment completed in only 10 iterations by leveraging existing knowledge (Figure 7), demonstrating the value of accumulated terrain understanding for repeated missions.

d) Goal-Free Exploration: Without predefined destinations, the robot selects points maximizing uncertainty reduction, rapidly expanding the known safe area (Figure 8, Figure 9). This demonstrates SRPGT's capability for general environmental mapping, useful for initial site characterization before goal-directed missions.

C. Comparison to Prior Gaussian Process-Based Planning

The framework proposed by Leininger et al. [8] combines Sparse Gaussian Processes (SGP) with RRT*-based motion planning. This approach offers global terrain reasoning and path planning capabilities using uncertainty-aware models. SRPGT distinguishes itself from Leininger et al. [8] through a number of architectural and operational design choices that make it more suitable for granular terrain exploration using local feedback.

- **Planning Strategy.** Leininger et al. [8]'s approach constructs global paths to the goal using RRT* over a probabilistic terrain cost map. This requires maintaining and updating a global terrain estimate and is most effective in structured or semi-known environments. In contrast, SRPGT takes a reactive planning approach that

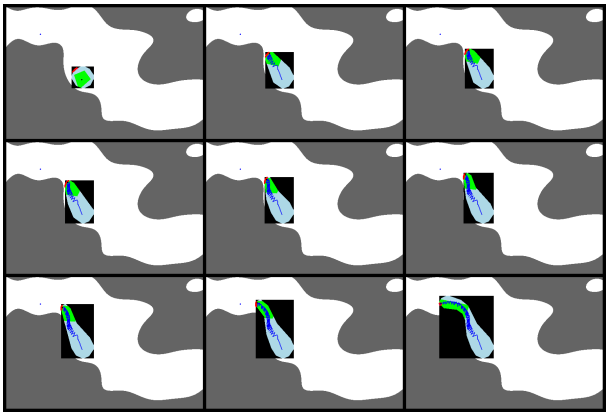

Fig. 5. Navigation around risk zone ($h = 1000$) over 9 intervals.

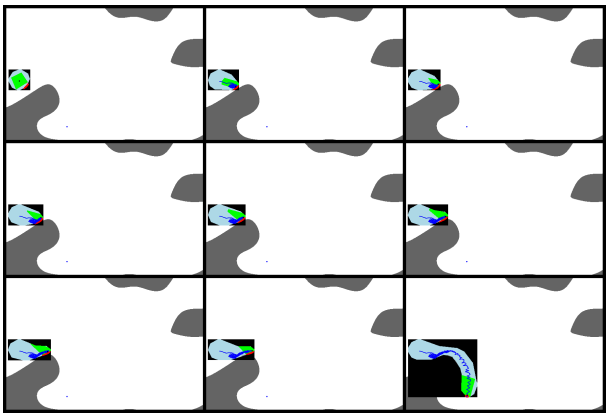

Fig. 6. Navigation around risk zone ($h = 500$) over 9 intervals.

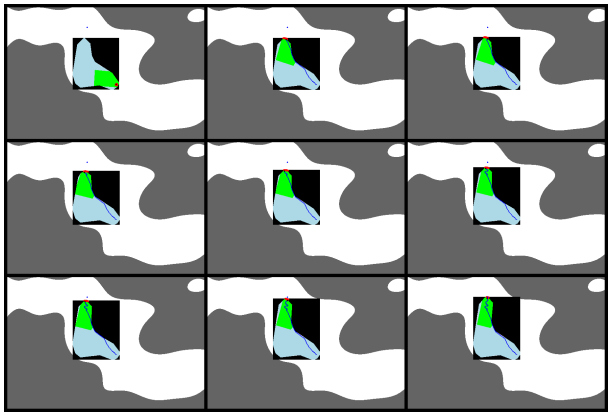

Fig. 7. Traversal across known safe area in 9 intervals.

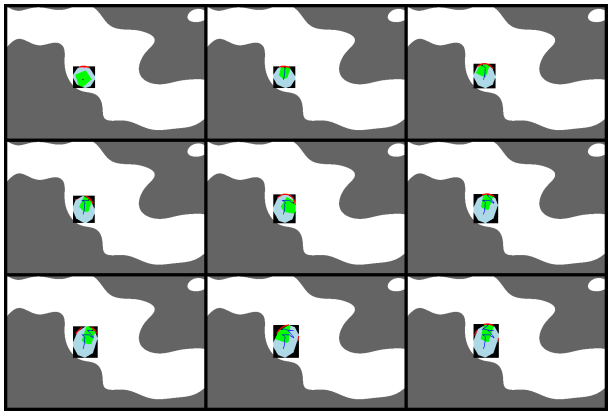

Fig. 8. First 9 frames of exploration task.

forgoes global path computation in favor of incremental, confidence-aware subgoal selection from a locally expanding safe set. This supports greater flexibility in dynamically evolving or unknown environments.

- **Uncertainty Handling.** While both methods utilize Gaussian Process models to estimate terrain risk and uncertainty, SRPGT directly incorporates the GP confidence bounds into its decision-making loop. The confidence intervals define both the safe and expander sets that determine feasible and informative next steps. In the SGP-RRT* formulation, uncertainty is treated as a scalar cost modulation rather than as a constraint mechanism to enforce safety under uncertainty.
- **Sensing Assumptions.** A critical difference lies in sensing modality. The SGP-RRT* planner assumes access to exteroceptive observations, such as visual or remote terrain features, to update the GP model. In contrast, SRPGT is built for scenarios in which such sensing may be unavailable or unreliable. It relies solely on proprioceptive feedback gathered through direct terrain interaction—making it particularly well-suited for planetary missions in dust-obscured or visually degraded environments.

Taken together, these distinctions highlight how SRPGT extends the capabilities of uncertainty-aware terrain modeling into domains where traditional planning pipelines fall short. The integration of proprioceptive sensing, real-time safe set expansion, and local control makes SRPGT more resilient in scenarios where structural assumptions break down.

D. Conclusions from Simulation Trials

The simulation experiments presented in this chapter validate the intended capabilities of SRPGT under a variety of terrain conditions and risk scenarios. The robot successfully performs uncertainty-aware navigation around hazardous terrain, reuses previously acquired safe knowledge to complete tasks more efficiently, and performs autonomous environmental mapping using only local proprioceptive measurements.

V. Future Work

Several key directions remain for enhancing the proposed method.

A. Adaptive Kernel Models

The current RBF kernel assumes spatial homogeneity with fixed length scales, which may be suboptimal for heterogeneous terrain. Future work could explore adaptive kernels [3] that dynamically adjust length scales based on local terrain variation. This would enable efficient exploration in simple

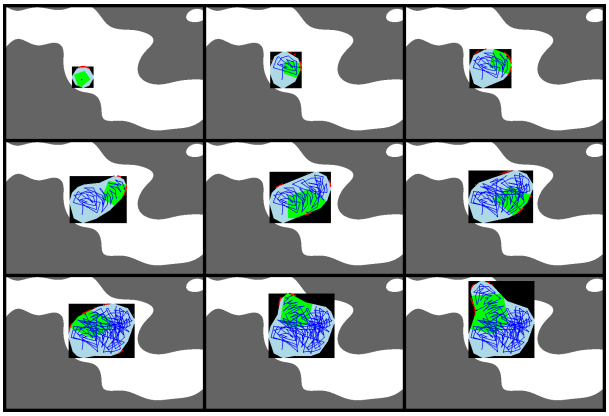

Fig. 9. Overall exploration progression over 9 intervals.

regions while maintaining caution in complex areas, reducing sampling requirements and improving safety.

B. Bayesian Planning for Coherent Exploration

The current greedy sampling approach can cause inefficient ”zig-zagging” when traversing uncertain terrain frontiers, as uncertainty reduction around each expander makes neighboring points temporarily most uncertain (Figure 10).

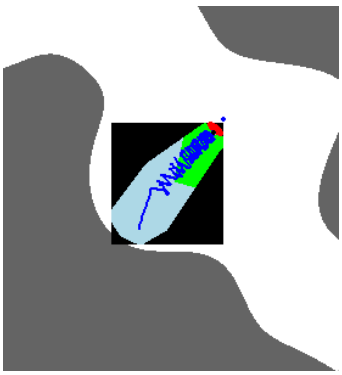

Fig. 10. ”Zig-Zag” behavior of the robot when searching in a straight line.

Incorporating belief-space planning frameworks like POMDPs could address this by considering long-term information gain rather than immediate uncertainty reduction. This would promote smoother, globally-directed exploration while maintaining local safety.

C. Integration with Exteroceptive Sensing

While SRPGT effectively evaluates terrain risk through proprioceptive interaction, it assumes all safe areas are obstacle-free. In practice, safe terrain may contain non-traversable obstacles detectable through vision, LiDAR, or depth sensors.

Future work could fuse proprioceptive terrain maps with geometric occupancy grids to distinguish between risky terrain and physical obstacles. The existing diffeomorphic mapping framework could incorporate hard geometric constraints, enabling hybrid planning where geometric planners filter feasible corridors while SRPGT selects among them based on proprioceptive risk.

This integration would transform SRPGT from a proprioceptive-only planner into a complete terrain-aware framework leveraging multiple sensing modalities for both geometric and mechanical safety.

VI. Limitations

While SRPGT achieves robust performance in proprioceptive terrain-aware navigation, several limitations constrain its broader applicability and motivate future extensions.

First, the current framework operates exclusively on proprioceptive input and does not incorporate exteroceptive sensing such as visual or LiDAR data. While this design choice enables operation in visually degraded environments, it also precludes early hazard detection and limits long-range planning capabilities. A fully effective navigation system would benefit from integrating exteroceptive modalities to detect geometric obstacles, anticipate terrain transitions, and guide sampling toward informative regions beyond immediate contact.

Second, terrain risk is modeled as a scalar field with a fixed threshold for safety classification. This abstraction does not account for complex mechanical behaviors such as anisotropy, layered substrates, or time-varying deformation properties, which may impact traversability.

Lastly, the framework assumes unobserved regions are unsafe by default, which, while conservative, can lead to slow exploration progress in large or mostly benign environments. Incorporating belief models or priors from exteroceptive maps could improve sampling efficiency while maintaining safety.

These limitations highlight key areas for future development, particularly toward integrating proprioceptive safety reasoning with exteroceptive perception and dynamic control to support robust, efficient navigation in complex, real-world terrain.

VII. Conclusion

This paper presented a novel navigation framework for proprioceptive terrain-aware exploration in granular environments, where conventional visual sensing is unreliable or unavailable. The system was designed to address the core challenges of uncertainty, deformability, and incomplete knowledge that arise in planetary terrain scenarios. By integrating Gaussian Process-based terrain modeling, confidence-guided safe set expansion, and diffeomorphic reactive control, SRPGT enables legged robots to safely and adaptively explore unknown environments using only local physical interaction.

Through simulation-based validation, SRPGT demonstrated three key capabilities: safely navigating around high-risk terrain, efficiently reusing previously learned safe paths, and rapidly expanding known terrain through exploratory behavior. Unlike prior approaches that rely on global visual maps or precomputed paths, SRPGT achieves safety-aware navigation using real-time proprioceptive measurements and confidence-aware planning, filling a critical gap in current planetary robotics literature.

This work contributes a unified framework for autonomous, local-information-driven exploration under risk constraints, opening new possibilities for robotic operation in visually degraded and mechanically unstable environments.

Acknowledgments

Thank you.

References

[1] Omur Arslan and Daniel Koditschek. Exact robot navigation using power diagrams. 05 2016. doi: 10.1109/ICRA.2016.7487090.

[2] Omur Arslan and Daniel E Koditschek. Sensor-based reactive navigation in unknown convex sphere worlds. *The International Journal of Robotics Research*, 38(2-3): 196–223, 2019. doi: 10.1177/0278364918796267. URL https://doi.org/10.1177/0278364918796267.

[3] Weizhe Chen, Roni Khardon, and Lantao Liu. Ak: Attentive kernel for information gathering, 2022. URL https://arxiv.org/abs/2205.06426.

[4] cubao. Concave hull. https://github.com/cubao/concave_hull, 2024. Accessed April 30, 2025.

[5] Santiago Garrido and Luis Moreno. *Mobile Robot Path Planning using Voronoi Diagram and Fast Marching*. 05 2015. ISBN 9781466686939. doi: 10.4018/978-1-4666-8693-9.

[6] Léonard Jaillet, Juan Cortés, and Thierry Siméon. Sampling-based path planning on configuration-space costmaps. *IEEE Trans. Robotics*, 26(4):635–646, 2010.

[7] Sertac Karaman and Emilio Frazzoli. Sampling-based algorithms for optimal motion planning. *Int. Journal of Robotics Research*, 30(7):846–894, 2011.

[8] Abe Leininger, Mahmoud Ali, Hassan Jardali, and Lantao Liu. Gaussian process-based traversability analysis for terrain mapless navigation, 2024. URL https://arxiv.org/abs/2403.19010.

[9] Arpan Losalka and Jonathan Scarlett. Benefits of monotonicity in safe exploration with gaussian processes. In *Proceedings of the 39th Conference on Uncertainty in Artificial Intelligence (UAI)*, pages 1304–1314, 2023. URL https://arxiv.org/abs/2211.01561.

[10] Justin N. Maki, David Gruel, Constance McKinney, Michael A. Ravine, Michael Morales, Douglas Lee, Richard Willson, David Copley-Woods, Michael Valentin, Thomas Goodsall, James McGuire, R. Greg Sellar, Jake A. Schaffner, Ken D. Supulver, Jennifer Tu, Andres Wong, Paul Brugarolas, Samad A. Hayati, Michael Kasper, Subash Reddy, and Jeff Umland. The mars 2020 engineering cameras and microphone on the perseverance rover: A next-generation imaging system for mars exploration. *Space Science Reviews*, 216(8): 1–48, 2020. doi: 10.1007/s11214-020-00765-9.

[11] Kasidit Muenprasitivej, Jesse Jiang, Abdulaziz Shamsah, Samuel Coogan, and Ye Zhao. Bipedal safe navigation over uncertain rough terrain: Unifying terrain mapping and locomotion stability, 2024. URL https://arxiv.org/abs/2403.16356.

[12] Randall Munroe. Spirit, 2009. URL https://xkcd.com/695/. Accessed: 2025-04-29.

[13] Michael Otte and Emilio Frazzoli. Rrt-x: Real-time motion planning/replanning for environments with unpredictable obstacles. In *Algorithmic Foundations of Robotics XI*, pages 461–478. Springer, 2015.

[14] J.-S Park and S.-J Oh. A new concave hull algorithm and concaveness measure for n-dimensional datasets. *Journal of Information Science and Engineering*, 29:379–392, 03 2013.

[15] Anthony Stentz. Optimal and efficient path planning for partially-known environments. In *Proc. IEEE Int. Conf. on Robotics and Automation (ICRA)*, pages 3310–3317, 1994.

[16] Alex Stephens, Matthew Budd, Michal Staniaszek, Benoit Casseau, Paul Duckworth, Maurice Fallon, Nick Hawes, and Bruno Lacerda. Planning under uncertainty for safe robot exploration using gaussian process prediction. *Autonomous Robots*, 48(18), 2024. doi: 10.1007/s10514-024-10172-6.

[17] Yanan Sui, Alkis Gotovos, Joel Burdick, and Andreas Krause. Safe exploration for optimization with gaussian processes. In Francis Bach and David Blei, editors, *Proceedings of the 32nd International Conference on Machine Learning*, volume 37 of *Proceedings of Machine Learning Research*, pages 997–1005, Lille, France, 07–09 Jul 2015. PMLR. URL https://proceedings.mlr.press/v37/sui15.html.

[18] Monisha Mushtary Uttsha, Cedric Le Gentil, Lan Wu, and Teresa Vidal-Calleja. Gaussian process distance fields obstacle and ground constraints for safe navigation, 2024. URL https://arxiv.org/abs/2410.17831.

[19] V. Vasilopoulos, G. Pavlakos, S. L. Bowman, J. D. Caporale, K. Daniilidis, G. J. Pappas, and D. E. Koditschek. Reactive Semantic Planning in Unexplored Semantic Environments Using Deep Perceptual Feedback. *IEEE Robotics and Automation Letters (RA-L)*, 5(3):4455–4462, 2020. doi: 10.1109/LRA.2020.3001496. URL https://doi.org/10.1109/ICRA.2016.7487090.

[20] Vasileios Vasilopoulos, Georgios Pavlakos, Karl Schmeckpeper, Kostas Daniilidis, and Daniel E. Koditschek. Reactive navigation in partially familiar planar environments using semantic perceptual feedback, 2021. URL https://arxiv.org/abs/2002.08946.